

CSI 4118: Computer Networks & Protocols (Fall 2025)

- **Professor**

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- **Lectures**

Tuesdays **10:00-11:30**
Thursdays **8:30-10:00**

- **Prerequisites**

SEG3155 (Communication and Networking)

- **Calendar Description**

Communication services, protocols and software. Details of layered protocol hierarchies. The transport, session, presentation and application layers. Fundamental concepts of computer network design. Computer network and communication protocol architectures.

- **Text Book**

William Stallings, *Data and Computer Communications (10th Ed.)*, Prentice Hall, 2013.

- **Recommended Reading**

- 1) Fred Halsall, *Computer Networking and the Internet (5th Ed.)*, Addison Wesley, 2005.
- 2) James F. Kurose, Keith W. Ross, *Computer Networking: A Top-Down Approach Featuring the Internet*, Addison Wesley, 2005.
- 3) Bahrouz A. Forouzan, *Data Communications and Networking (4th Ed.)*, McGraw Hill, 2007.
- 4) Douglas E. Comer, *Computer Networks and Internets with Internet Applications (4th Ed.)*, Prentice Hall, 2004.
- 5) Andrew S. Tanenbaum, *Computer Networks (4th Ed.)*, Prentice Hall, 2003.

- **Marking Scheme**

Assignments	20%
Test #1 (Thursday, Oct. 2)	15%
Test #2 (Thursday, Nov. 6)	15%
Final	50%

- **Syllabus**

1. Review of computer networks and general networking concepts.
2. Internet protocols.
3. Internetwork operation (OSPF, RIP, BGP, IGMP, DVMRP, PIM, MPLS).
4. Transport layer protocols (TCP and UDP).
5. Communication architectures, socket programming.
6. Network performance measurement and management issues.
7. Application layer protocols (DNS, FTP, SMTP, HTTP, SNMP, etc.).
8. Network security.
9. Multimedia networking (RTP, RTCP, RTSP, VoIP, SIP, H.323, RSVP, etc.).
10. Peer-to-peer networking, Mobile IP, Software-defined networking.
11. Ad hoc routing protocols (AODV, OLSR, DSDV, DSR, etc.)